BANK MARKETING SUBSCRIPTION PREDICTION: A COMPARISON OF TRADITIONAL MACHINE LEARNING AND A NEURAL NETWORK

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Course: **Introduction to Machine Learning**

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GitHub: **https://github.com/armstrong-codes/bank-marketing-classification.git**

Dataset and Folders:
https://drive.google.com/drive/folders/1a0Z_xTWt4xHZLUio5KmDdNXLWWqpCnRa?usp=drive_link

Presentation: **https://youtu.be/pTLh122W6bA**

Abstract

This project investigates supervised classification methods for predicting whether a customer contacted during a bank marketing campaign subscribes to a term deposit. The study uses the UCI Bank Marketing dataset and compares traditional classifiers—Logistic Regression, Random Forest, and Histogram-based Gradient Boosting—with a tuned Random Forest and a feed-forward neural network implemented using TensorFlow/Keras. The workflow includes exploratory analysis, preprocessing of numeric and categorical features, feature engineering, stratified data splitting, hyperparameter tuning, model training, and evaluation with metrics suited to binary classification with an imbalanced target. On the held-out test set, the tuned Random Forest achieved 86.18% accuracy, 42.62% precision, 52.46% recall, an F1-score of 47.03%, ROC-AUC of 0.7895, and average precision of 0.4519. The neural network achieved 89.37% accuracy and 61.76% precision, but its recall was 23.83% and its F1-score was 34.39%; its ROC-AUC was 0.7882 and average precision was 0.4373. Thus, the neural network produced higher accuracy and precision at the selected threshold, while the Random Forest identified more actual subscribers and achieved a higher F1-score. The findings demonstrate why accuracy alone is insufficient for model selection when classes are imbalanced. The preferred model depends on the relative cost of missing potential subscribers and contacting customers who do not subscribe.

Index Terms

Bank marketing, binary classification, Random Forest, Logistic Regression, gradient boosting, neural network, class imbalance, ROC-AUC, precision, recall.

I. Introduction

Marketing teams must decide which customers should receive promotional contact. A bank may promote term deposits through telephone campaigns, but contacting every customer can consume staff time and create unnecessary interruptions. Classification models can estimate whether a customer is likely to subscribe and help prioritize outreach. These predictions should support rather than replace human judgment because model outputs are uncertain and may reflect patterns or biases in historical data.

This project treats subscription as a binary classification problem. The target variable y records whether the customer subscribed (yes) or did not subscribe (no). The dataset combines customer attributes, contact information, and campaign-history variables, allowing comparison of algorithms with different modelling assumptions.

The objectives are to: (1) inspect and preprocess the dataset; (2) compare traditional classifiers; (3) tune a Random Forest and train a TensorFlow/Keras neural network using tf.data; and (4) compare their performance using several metrics. The research question is: how do traditional machine-learning classifiers and a feed-forward neural network compare when predicting term-deposit subscription, particularly when the positive class is less common than the negative class?

II. Related Work and Technical Background

Moro, Rita, and Cortez introduced the Bank Marketing dataset as a resource for studying direct-marketing campaign outcomes [1]. It includes customer information and campaign interaction variables. One important methodological issue is prediction timing: a variable recorded only after a call can make an offline model appear more useful than it would be before the call.

Logistic Regression is a common baseline for binary classification. It estimates the probability of a positive outcome from a weighted combination of input features. Its relative simplicity makes it useful for benchmarking, although its linear decision boundary may not capture every nonlinear relationship.

Random Forest combines predictions from multiple decision trees trained with randomized samples and feature subsets [2]. This ensemble approach can model nonlinear relationships and reduce the instability of a single tree. Its performance depends on settings such as tree depth, number of trees, and minimum sample sizes. Histogram-based Gradient Boosting builds trees sequentially, with later trees correcting errors from earlier ones. It can capture nonlinear relationships but also requires suitable model complexity and tuning.

A feed-forward neural network transforms input features through learned weights and nonlinear activation functions. The Keras Sequential API represents a linear stack of layers [3]. A sigmoid output can represent the estimated probability of the positive class. The network's behaviour depends on architecture, optimizer, learning rate, batch size, and stopping criteria.

Accuracy can be misleading when one class dominates. Precision is the proportion of positive predictions that are correct; recall is the proportion of actual positives identified; and F1 is the harmonic mean of precision and recall. ROC-AUC measures ranking performance over thresholds, while average precision summarizes precision across recall levels and can be informative when positive examples are relatively uncommon.

III. Dataset and Preprocessing

A. Dataset Source and Target

The project uses the Bank Marketing dataset from the UCI Machine Learning Repository [1]. The notebook is designed for bank-full.csv, which is expected to contain 45,211 records and 17 columns. Confirm the exact filename and dimensions from the executed notebook before submitting, particularly if another dataset version was loaded. The target y indicates whether a customer subscribed to a term deposit. The notebook maps no to 0 and yes to 1, making yes the positive class.

B. Exploratory Analysis

Exploratory analysis checks dataset dimensions, data types, target distribution, missing values, duplicates, and feature distributions. These checks identify data-quality issues and reveal whether target classes are balanced. A model can obtain apparently high accuracy by predicting the majority class, even when it fails to identify the less common class.

Categorical variables describe groups such as occupation, marital status, education, contact type, month, and previous campaign outcome. Numeric variables include quantities such as age and campaign-contact counts. Values such as unknown may represent unavailable information and should be interpreted cautiously.

C. Feature Engineering and Leakage

The notebook creates a previously_contacted indicator from campaign-history information. Such a feature may make an existing pattern easier for the model to use, but should only be included if it is available at the time predictions are intended to be made.

The duration variable is excluded because the intended scenario is to predict subscription before the call outcome is known. Call duration is only available after a call and may be strongly associated with the target. Including it in a pre-call model would leak information from the future and could overstate realistic performance. This exclusion makes the use case more realistic, although it may reduce apparent predictive performance.

D. Data Splitting and Preprocessing

The workflow plans a stratified 70/15/15 split for training, validation, and test data. Stratification aims to preserve class proportions across subsets. Training data are used to fit model parameters; validation data support comparison and selection; and the test set provides a final evaluation after decisions are made. Repeatedly using the test set for tuning would weaken its independence.

Numeric features are imputed where necessary and scaled. Categorical features are imputed and one-hot encoded, representing categories without imposing a false numerical ordering. Preprocessing transformations should be fitted on training data only and then applied unchanged to validation and test data to reduce information leakage.

IV. Methodology

A. Traditional Classifiers

The traditional model comparison includes Logistic Regression, Random Forest, and HistGradientBoosting. Logistic Regression provides a linear probabilistic baseline; Random Forest averages the predictions of multiple trees; and HistGradientBoosting builds a sequence of trees that correct previous errors. Their validation performance was compared using precision, recall, F1-score, ROC-AUC, and average precision.

The supplied validation chart suggests that HistGradientBoosting and Random Forest had precision around 0.64–0.66 and recall around 0.22–0.24, while Logistic Regression had precision around 0.27 and recall around 0.63. These are approximate values read from the chart. Replace them with exact values from the notebook's metric table before submission.

B. Random Forest Tuning

GridSearchCV was used to evaluate candidate Random Forest configurations with cross-validation, selecting according to F1-score. F1 is relevant where both precision and recall matter. The best visible mean cross-validation score is approximately 0.4618 with standard deviation 0.0091.

Some tuning candidates achieved near-perfect training scores but lower cross-validation scores. This gap is a warning sign of overfitting: a model may fit the training examples extremely closely without generalizing as well to unseen observations. Cross-validation scores are more useful than training scores alone for selecting a configuration.

C. Neural Network

The notebook implements a feed-forward neural network with TensorFlow/Keras, using a Sequential architecture and a tf.data input pipeline. The documented design uses dense hidden layers of 128 and 64 units with ReLU activation, dropout regularization, and a sigmoid output for binary classification. The input pipeline supplies batches of preprocessed features and labels.

The network is trained using binary cross-entropy, and early stopping monitors validation performance to limit unnecessary training. Before submission, copy the exact optimizer, learning rate, batch size, dropout rate, maximum epochs, and early-stopping patience from the code.

D. Evaluation Metrics

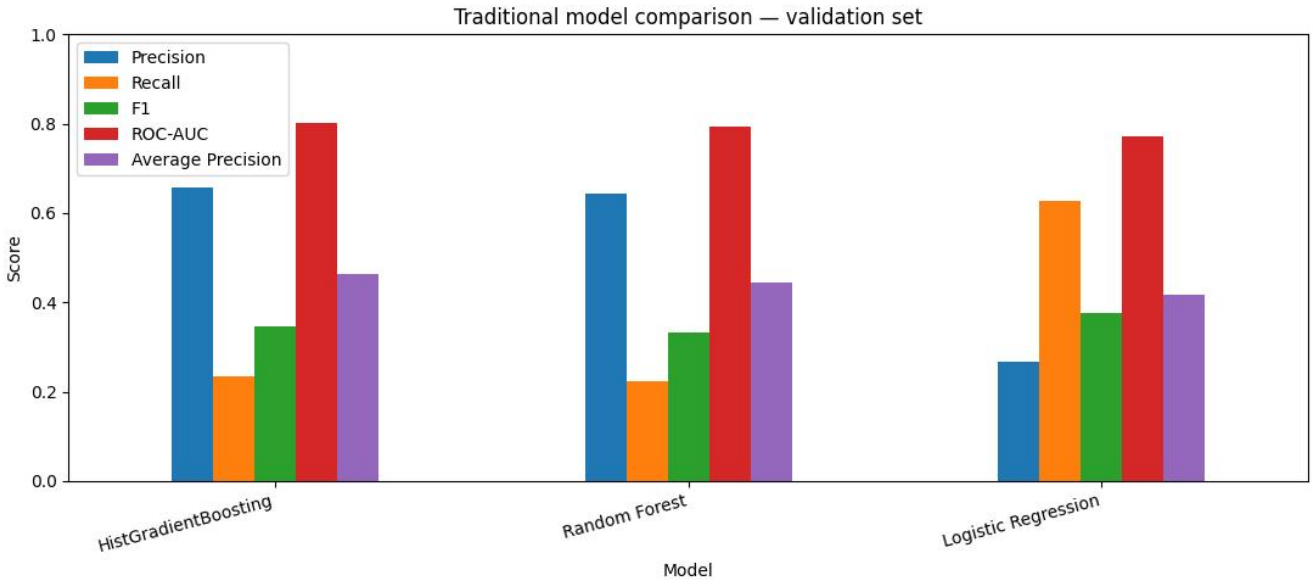
Let TP be true positives, TN true negatives, FP false positives, and FN false negatives. Precision = TP/(TP+FP); recall = TP/(TP+FN); accuracy = (TP+TN)/(TP+TN+FP+FN); and F1 = 2×precision×recall/(precision+recall). ROC-AUC summarizes ranking ability across thresholds, while average precision summarizes precision across recall levels. Confusion matrices reveal the counts and types of errors at a chosen threshold.

Because the target is imbalanced, the analysis considers precision, recall, F1, ROC-AUC, and average precision alongside accuracy. ROC curves show how the true-positive rate changes with the false-positive rate across thresholds. Learning curves help assess training progress and differences between training and validation performance.

V. Experimental Results

A. Traditional Validation Comparison

The validation comparison showed that no model dominated every metric. HistGradientBoosting and Random Forest were more selective in positive predictions, with higher precision but lower recall. Logistic Regression identified a larger share of actual subscribers, but produced more false positives. The choice reflects a trade-off: a broad outreach list can capture more potential subscribers, while a selective list can reduce unnecessary contacts.



B. Test-Set Metrics

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Avg. Precision
Tuned Random Forest	0.8618	0.4262	0.5246	0.4703	0.7895	0.4519
Neural Network	0.8937	0.6176	0.2383	0.3439	0.7882	0.4373

The neural network achieved higher test accuracy (89.37%) and precision (61.76%). It classified more observations correctly overall and a greater proportion of its positive predictions were actual subscribers. However, its recall was 23.83%, so it detected fewer than one quarter of actual subscribers at the selected threshold. Its F1-score was 34.39%, lower than the Random Forest's 47.03%.

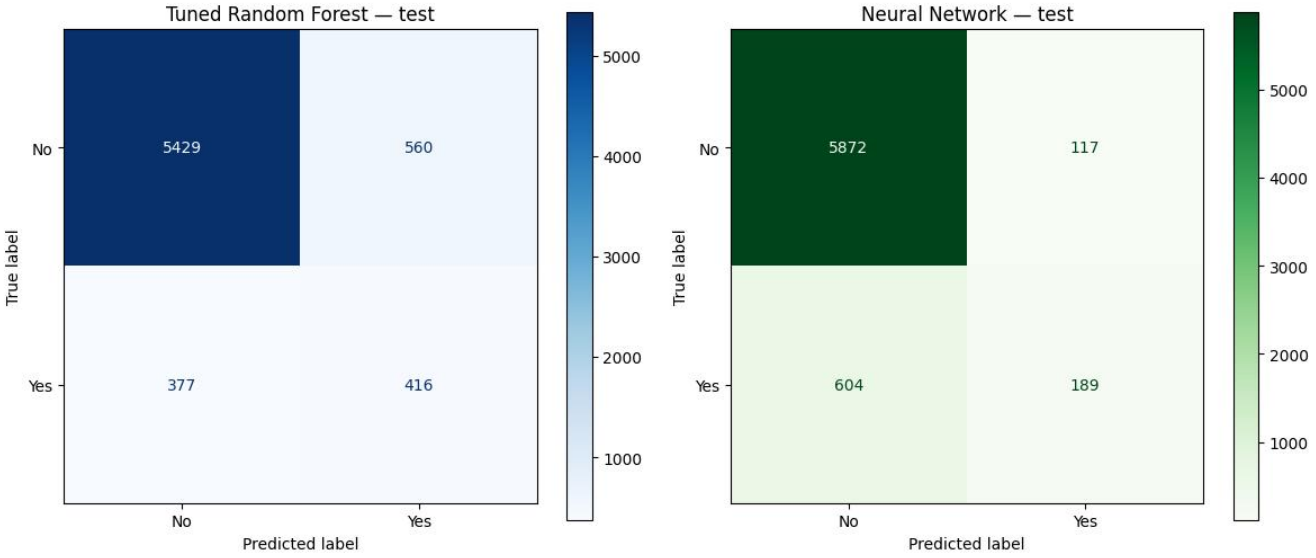
The tuned Random Forest achieved recall of 52.46% and F1-score of 47.03%, identifying a larger proportion of actual subscribers and achieving a better balance of precision and recall. Its ROC-AUC of 0.7895 was only slightly higher than the neural network's 0.7882, indicating similar ranking performance rather than a decisive difference. Average precision was also slightly higher for the Random Forest (0.4519 compared with 0.4373).

These results show why accuracy alone is not a sufficient selection criterion. In an imbalanced dataset, a model may classify the majority class correctly and still miss many positive cases. The preferred model depends on the bank's objective and the cost of different errors. If missing potential subscribers is costly, the Random Forest is more effective at the current threshold. If precision is the priority, the neural network is more selective, but its low recall is a substantial limitation.

C. Confusion Matrices

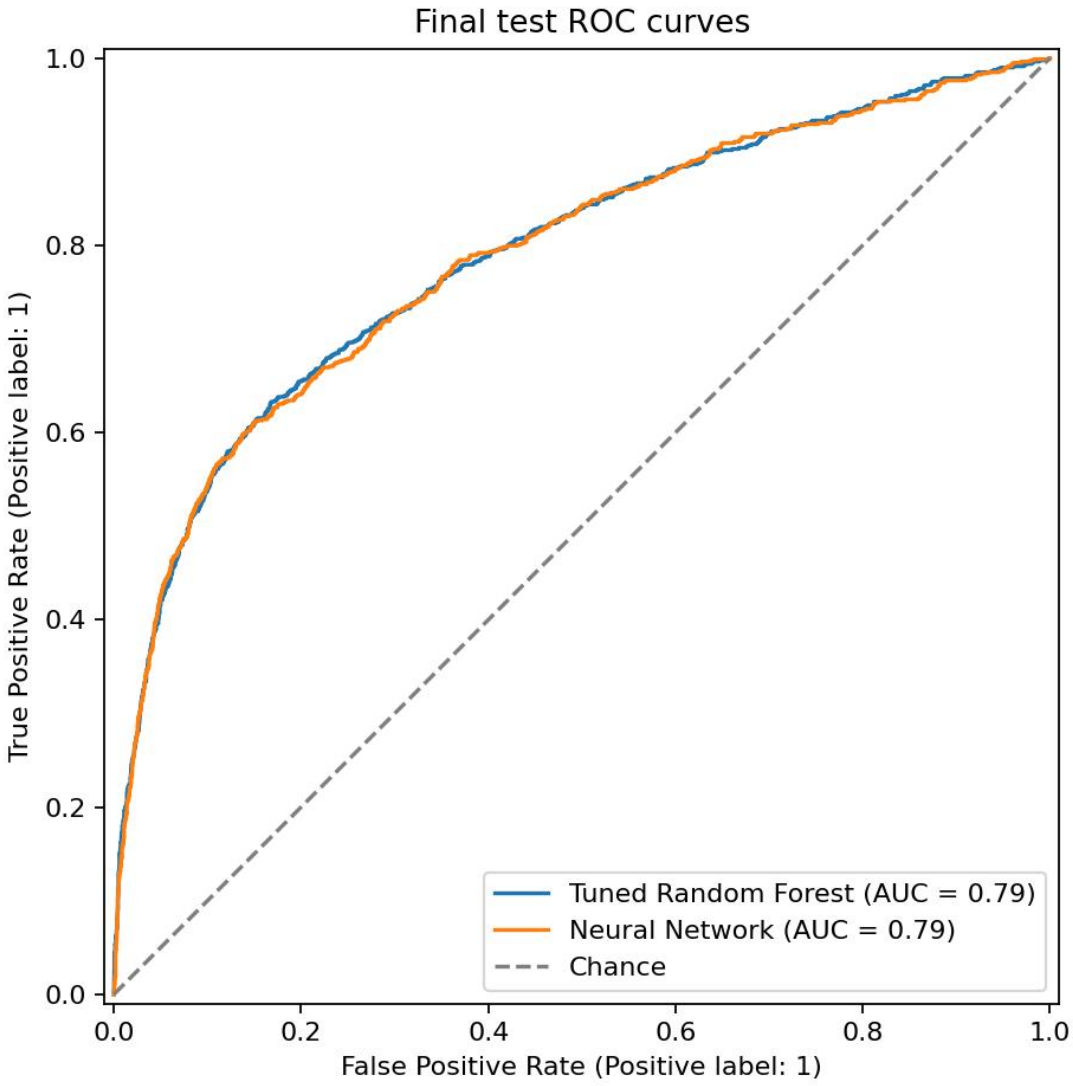
The Random Forest test confusion matrix contains 5,429 true negatives, 560 false positives, 377 false negatives, and 416 true positives. It correctly classifies many non-subscribers and identifies 416 of the 793 actual subscribers, while missing 377 subscribers.

The neural-network confusion matrix contains 5,872 true negatives, 117 false positives, 604 false negatives, and 189 true positives. It produces far fewer false-positive predictions but misses many more actual subscribers. This error pattern explains its relatively high precision and low recall.



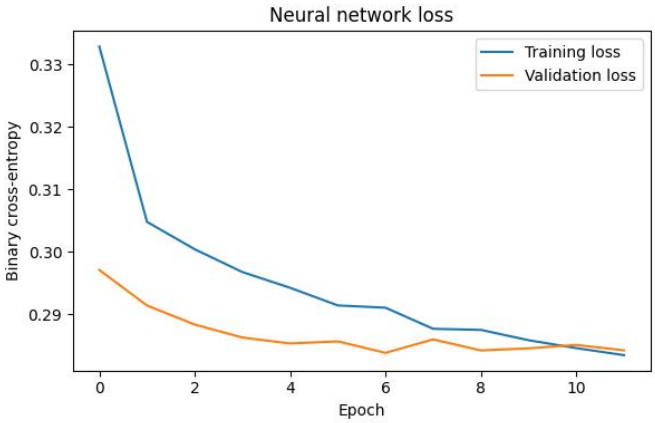
D. ROC Curves

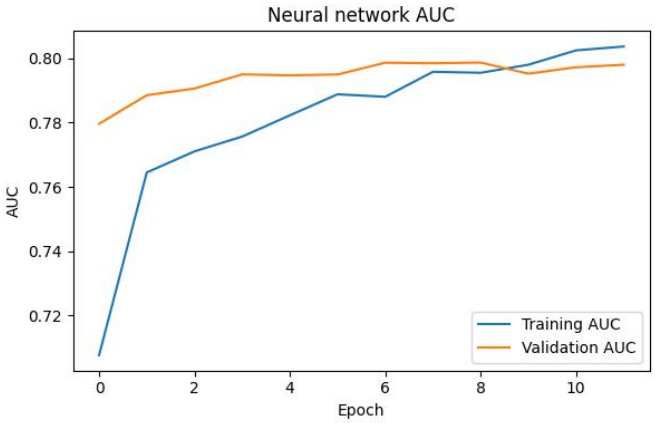
The final ROC curves are close to each other, with AUC values near 0.79. Both curves lie above the diagonal chance line, indicating better ranking than random ordering. However, ROC-AUC does not specify the best operational threshold and does not guarantee high recall or precision at the default threshold.



E. Neural-Network Learning Curves

The displayed training loss decreases over epochs, while validation loss remains relatively stable with small fluctuations. Training AUC increases and validation AUC remains near 0.79–0.80. The displayed curves do not show a large widening gap between training and validation AUC, suggesting that the network learns useful patterns without an obvious large generalization gap in these epochs. However, stable curves do not guarantee a suitable classification threshold or high recall.





VI. Discussion

A. Interpretation

The results do not show that the neural network outperformed the tuned Random Forest across all metrics. It achieved higher accuracy and precision, but the Random Forest achieved higher recall, F1-score, ROC-AUC, and average precision. The small AUC difference suggests that both models rank customers similarly overall, even though their predictions at the selected threshold differ substantially.

One possible explanation is that the models produce different probability distributions and therefore make different classifications at the default threshold. The neural network may assign lower probabilities to many actual subscribers, causing them to be classified as No, while the Random Forest labels more cases as Yes. Threshold analysis and probability calibration would be needed to examine this explanation further.

B. Business Implications

A false negative is a potential subscriber not selected by the model; a false positive is a customer selected for outreach who does not subscribe. These errors have different costs. If campaign capacity allows more calls and the bank values reaching potential subscribers, the Random Forest's higher recall may be preferable. If outreach resources are limited and avoiding unnecessary calls is more important, the neural network's higher precision may be useful, but its low recall means that many potential subscribers would be missed.

A real deployment decision should consider call costs, the expected value of a subscription, available campaign capacity, and customer-experience effects. A threshold can be chosen using validation data to meet a defined business objective, such as achieving a minimum recall while keeping false positives within operational capacity.

C. Limitations

The dataset represents historical campaigns in a particular banking context and may not reflect current customer behaviour, economic conditions, or other institutions. The target is imbalanced, and metrics depend on threshold choice and class proportions. The customers contacted historically may not represent all potential customers, creating selection effects.

Excluding duration helps avoid leakage for a pre-call prediction scenario, but every feature should be checked for availability at the intended decision time. Results are based on a held-out split and may vary across samples. The analysis also does not establish causality: a customer predicted to subscribe is not necessarily someone whose decision can be changed by a marketing call.

D. Future Work

Further experiments could select classification thresholds using validation data, test class weighting, evaluate probability calibration, and use repeated stratified cross-validation. Feature-importance analysis could help explain predictions, but associations should not be treated as causal effects. Temporal validation would be useful if campaign dates are available, by training on earlier campaigns and testing on later ones. A duration-inclusive model could be studied as a post-call benchmark, but should be clearly distinguished from a realistic pre-call predictor.

E. Quantitative Error Analysis

The confusion matrices provide a more operational view of the test results than the summary metrics alone. The test set contains 6,782 observations, including 793 customers who subscribed and 5,989 who did not. Therefore, subscribers account for approximately 11.7% of the test set. This imbalance is important when interpreting accuracy: a classifier that predicts the majority class for every customer would already achieve about 88.3% accuracy, despite identifying no subscribers at all. The neural network's accuracy of 89.37% is only modestly above that simple majority-class reference, so its accuracy should not be interpreted as evidence that it is the strongest model for the business objective.

The tuned Random Forest correctly identifies 416 of the 793 actual subscribers and misses 377. Its recall of 52.46% means that it finds slightly more than half of the positive cases in the test set. It also produces 560 false positives: customers predicted to subscribe who in fact did not subscribe. Among the 976 customers predicted positive by this model (416 true positives plus 560 false positives), about 42.6% subscribed, which matches its precision. This could be useful if the bank is willing to contact a wider group to reach more potential subscribers.

The neural network identifies 189 actual subscribers and misses 604. Its 23.83% recall means it detects fewer than one in four positive cases at the chosen threshold. In contrast, it produces only 117 false positives. Of the 306 customers predicted positive (189 true positives plus 117 false positives), approximately 61.8% subscribed. This is a more selective list, but the selectivity comes with a substantial loss in coverage: compared with the Random Forest, the network finds 227 fewer actual subscribers on the same test set. The comparison illustrates that precision and recall describe different dimensions of performance rather than competing versions of the same measure.

F. Threshold Choice and Operational Trade-offs

Both models produce scores that can be converted into class predictions using a decision threshold. A common default is 0.5, but this value is not automatically optimal for a marketing campaign. Lowering a threshold generally allows more customers to be classified as likely subscribers. This can increase recall, but it can also increase the number of false positives and the number of calls required. Raising the threshold usually makes the model more selective, often improving precision while reducing recall. The observed confusion matrices show that the two models make different trade-offs at their current thresholds, even though their ROC-AUC values are similar.

A more complete threshold experiment should use the validation set rather than the final test set. For each candidate threshold, the analyst can calculate precision, recall, F1-score, and the number of customers selected for outreach. A threshold could then be chosen to meet a practical requirement, such as identifying at least a specified proportion of likely subscribers while keeping the call list within the bank's capacity. After the threshold is selected, the test set should be used once for a final evaluation. Repeatedly choosing thresholds based on test performance would make the final estimate optimistic because the test set would become part of the model-selection process.

G. Understanding ROC-AUC and Average Precision

The tuned Random Forest has a ROC-AUC of 0.7895, while the neural network has a ROC-AUC of 0.7882. The difference is 0.0013, which is very small. These results suggest that the models have similar overall ability to rank positive examples above negative examples across possible thresholds. They do not establish that the models are equivalent, and no statistical significance test was performed. The practical difference in their confusion matrices can arise because a single threshold converts continuous scores into yes-or-no decisions. ROC-AUC assesses ranking across thresholds; precision, recall, and the confusion matrix describe performance at a particular threshold.

Average precision is 0.4519 for the Random Forest and 0.4373 for the neural network. This metric summarizes the precision-recall relationship and is particularly informative when the positive class is relatively uncommon. The Random Forest's higher average precision is consistent with its slightly stronger performance on this summary measure, although the difference is not large. Neither metric alone answers whether the model is profitable to deploy. A bank would still need to estimate the costs of calling customers, the value of successful subscriptions, the number of calls that staff can make, and the effect of repeated or unwanted contact.

H. Model Complexity, Tuning, and Generalization

The comparison also demonstrates why model complexity should be evaluated together with generalization. Logistic Regression provides a relatively simple baseline and can be easier to explain, but its linear form may not capture complex interactions among customer characteristics and campaign history. Random Forest and Histogram-based Gradient Boosting can represent nonlinear relationships and interactions without requiring the analyst to specify each interaction manually. A neural network can also learn nonlinear combinations of input features, but its performance depends on architecture, optimization, regularization, preprocessing, and training choices. Greater complexity does not guarantee better results on unseen data.

GridSearchCV supports a more systematic Random Forest comparison by evaluating candidate configurations using cross-validation. Selecting according to F1-score aligns the tuning objective with a balance between precision and recall. The visible mean cross-validation F1-score of approximately 0.4618, with a standard deviation of about 0.0091, indicates that the selected configuration achieved a similar score across the evaluated folds, although the precise interpretation depends on the folds and search space used. The exact parameter combination should be reported from the notebook output. It is also important to distinguish cross-validation performance from final test performance: the former guides model selection, while the latter estimates performance on held-out observations.

I. Interpreting the Neural-Network Learning Curves

The learning curves show training loss decreasing over epochs, while validation loss remains comparatively stable and the training and validation AUC values approach approximately 0.80. This pattern suggests that optimization improved the fit to the training data without a large visible divergence in AUC. However, it does not prove that the model is free of overfitting or that its classification decisions are appropriate. Loss and AUC measure different properties: loss evaluates the quality of predicted probabilities against labels, while AUC evaluates ranking. A model can show a stable AUC and still have an unsatisfactory recall at the chosen classification threshold.

The neural network's test accuracy and precision are relatively high, yet its recall is low. This combination suggests that the model predicts the positive class conservatively at the operating threshold. It would be useful to inspect the distribution of predicted probabilities for positive and negative cases and test alternative thresholds using validation data. Probability calibration could also be examined to determine whether predictions such as 0.7 correspond to an approximately 70% observed subscription rate among comparable cases. These are proposed follow-up analyses, not results established by the current experiment.

J. Validity, Reproducibility, and Responsible Use

Several steps support the validity of the experiment. The target is defined consistently, the split is stratified, preprocessing is designed to be fitted on training data, and the duration variable is excluded for a pre-call prediction scenario. These choices reduce specific risks of leakage and misleading evaluation. Nevertheless, a random stratified split does not necessarily reproduce a future campaign, where customer behaviour, market conditions, and campaign practices may

change over time. If date or campaign-order information is available, testing on later observations after training on earlier ones would provide a stronger assessment of performance under temporal change.

Reproducibility requires more than sharing the final metric values. The notebook should record the dataset version, random seeds, software dependencies, split sizes, preprocessing steps, selected hyperparameters, and training configuration. Saved model artifacts should be accompanied by the preprocessing components required to transform new data in the same way as training data. The report's tables and plots should be generated from the final notebook run so that the narrative cannot accidentally describe an earlier experiment. If results change after rerunning the notebook, the final report should use the outputs from the version actually submitted.

Finally, these models should be treated as decision-support tools rather than as definitive judgments about individual customers. Historical data may reflect past marketing choices and therefore may not represent customers who were never contacted. Unknown or incomplete feature values can also carry information about data collection practices rather than customer preferences. Before operational use, the bank should assess privacy requirements, customer-contact rules, subgroup performance, and the possibility of uneven error rates across groups. Predictive association also does not establish that contacting a customer will cause a subscription. A prospective campaign test would be needed to estimate the actual effect of model-guided targeting.

VII. Conclusion

This project compared traditional machine-learning classifiers and a TensorFlow/Keras neural network for predicting term-deposit subscription. The workflow included data inspection, preprocessing, feature engineering, traditional-model comparison, Random Forest tuning, neural-network training, and evaluation with multiple metrics.

On the test set, the neural network achieved 89.37% accuracy and 61.76% precision, but only 23.83% recall and a 34.39% F1-score. The tuned Random Forest achieved 86.18% accuracy, 42.62% precision, 52.46% recall, a 47.03% F1-score, ROC-AUC of 0.7895, and average precision of 0.4519. Both models had ROC-AUC near 0.79. The principal conclusion is that model choice should reflect the intended use and the costs of different errors, not accuracy alone. Threshold selection, robustness testing, and business-cost analysis are logical next steps.

Before submission, replace approximate values and placeholders with verified outputs from the final notebook run. The added analysis interprets the metrics already present in this report; it does not claim that the proposed threshold, calibration, temporal-validation, or subgroup experiments have been conducted.

References

[1] S. Moro, P. Rita, and P. Cortez, “Bank Marketing,” UCI Machine Learning Repository, 2014. doi: 10.24432/C5K306. [Online]. Available: https://archive.ics.uci.edu/dataset/222/bank+marketing

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